



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, March 5, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

Course introduction

Matteo Cianchetti - matteo.cianchetti@santannapisa.it



Structure of the course

TOPIC I – New approaches in robotics

- Soft Robotics
- Embodiment and Morphological Computation
- Bioinspiration and Biomimetics

TOPIC II – Actuation Technologies

- PE – PiezoElectric Actuators
- FFA – Flexible Fluidic Actuators
- SMA – Shape Memory Alloys
- EAP – Electro Active Polymers
- JT – Jamming Transition
- ERF-MRF – Electro- Magneto-Rheological Fluids

TOPIC III – Nonlinear analysis through FEM

- Fundamentals of FEM
- Nonlinear FEM analysis
- ANSYS

Seminar on "Soft Sensing"
by Dr. Maselli

Theory and Hands-on sessions

Seminar on "soft robotic artificial heart"
by Dr. Lorenzon

Theory and workshop sessions



References for the course

Topic I – New approaches in robotics

- “Soft Robotics”, *Cecilia Laschi*
- Slides provided through the website
- Referenced papers and books

Matteo Cianchetti
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Topic II – Actuation Technologies

- Slides provided through the website
- Referenced papers and books
- “Emerging Actuator Technologies – A Micromechatronic Approach”, *José L. Pons*

Topic III – Nonlinear analysis through FEM

- First 3 chapters of the book: “Concepts and Applications of Finite Element Analysis”, 4th Edition, *Robert D. Cook, David S. Malkus, Michael E. Plesha, Robert J. Witt*
- Slides provided through the website
- ANSYS software: Room 2 or student version freely available at:

<https://www.ansys.com/academic/free-student-products>

three-button
scroll mouse!

Office hours
by appointment

Exam (by appointment)
Oral: - theory on Topic I & II
- live simulation with ANSYS



Timetable - current

	gio 5/3	ven 6/3
Tutto il giorno		
08:30	P 08:30 - 11:30 BIOINSPIRED AND SOFT ROBOTICS - SOFT ROBOTICS TECHNOLOGIES BIONICS ENGINEERING (1) M. CIANCHETTI Lezioni svolte presso SSSUP S.Anna - Pontedera	
09:30		P 09:30 - 12:30 NEUROMORPHIC ENGINEERING BIONICS ENGINEERING (1) C. ODDO, A. MAZZONI Lezioni svolte presso SSSUP S.Anna - Pontedera
10:30		
11:30		
12:30		
13:30		
14:30	P 14:30 - 16:30 DATA MINING AND MACHINE LEARNING - ADVANCED DATA MINING AND MACHINE LEARNING ARTIFICIAL INTELLIGENCE AND DATA ENGINEERING (1) M. CIANCHETTI	P 14:30 - 16:30 BIOINSPIRED AND SOFT ROBOTICS - SOFT ROBOTICS TECHNOLOGIES BIONICS ENGINEERING (1) M. CIANCHETTI
15:30		
16:30	P 16:30 - 18:30 ELECTRONICS FOR BIONICS ENGINEERING BIONICS ENGINEERING (1) Ing A11 P. DINI, D. ROSSI	
17:30		



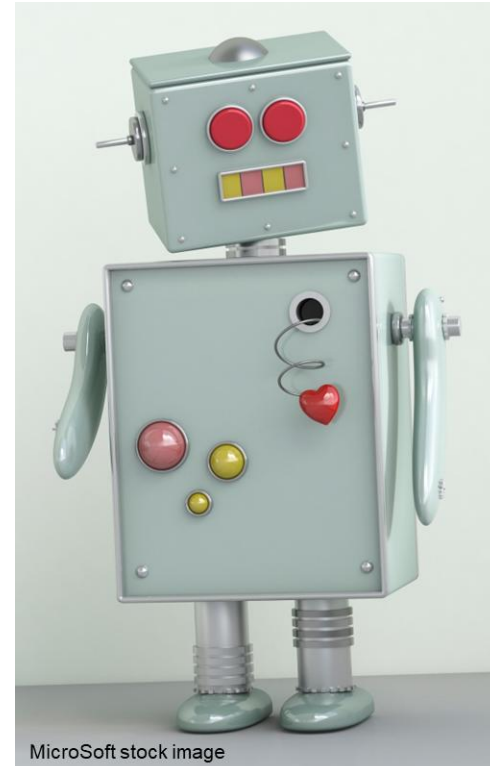
Timetable - proposal

	gio 5/3	ven 6/3
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08:30		
09:30	P 09:30 - 12:30 BIOINSPIRED AND SOFT ROBOTICS - SOFT ROBOTICS TECHNOLOGIES BIONICS ENGINEERING (1) M. CIANCHETTI Lezioni svolte presso SSSUP S.Anna - Pontedera	P 09:30 - 12:30 NEUROMORPHIC ENGINEERING BIONICS ENGINEERING (1) C. ODDO, A. MAZZONI Lezioni svolte presso SSSUP S.Anna - Pontedera
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13:30		
14:30	P 14:30 - 16:30 DATA MINING AND MACHINE LEARNING - ADVANCED DATA MINING AND MACHINE LEARNING ARTIFICIAL INTELLIGENCE AND DATA ENGINEERING (1) M. CIANCHETTI	P 14:30 - 16:30 BIOINSPIRED AND SOFT ROBOTICS - SOFT ROBOTICS TECHNOLOGIES BIONICS ENGINEERING (1) M. CIANCHETTI
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17:30		



A “Robot” Definition First

“ A robot is an autonomous system that exists in the physical world, can sense its environment, and can act on it to achieve some goals.



Matarić, Maja J. 2007. *The Robotics Primer*. Intelligent Robotics and Autonomous Agents Series. Cambridge, Mass: MIT Press.



A “Robot” Definition First

“

A robot is an

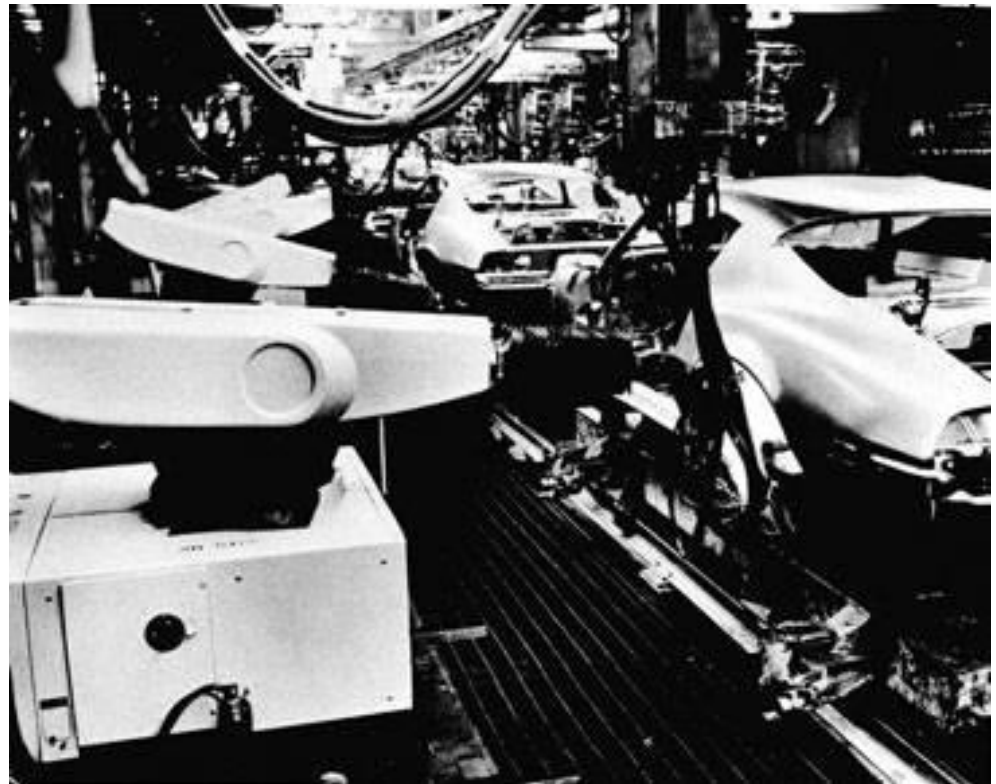
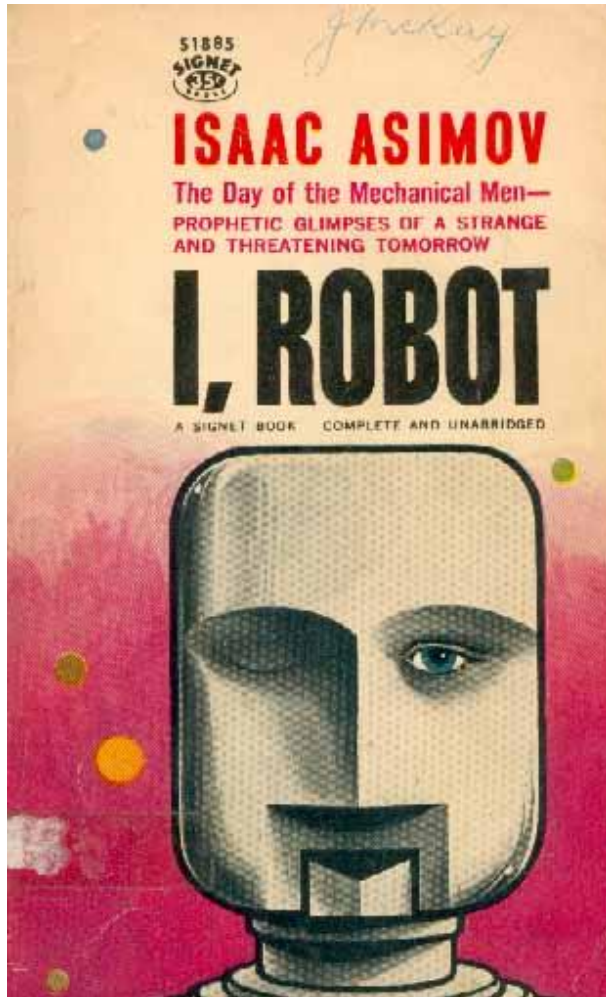
autonomous system that
exists in the **physical** world,
can **sense** its environment,
and can **act** on it
to achieve some **goals**.

≠ teleoperated systems
≠ digital agents
has sensors
has actuators
has a controller

Matarić, Maja J. 2007. *The Robotics Primer*. Intelligent Robotics and Autonomous Agents Series. Cambridge, Mass: MIT Press.



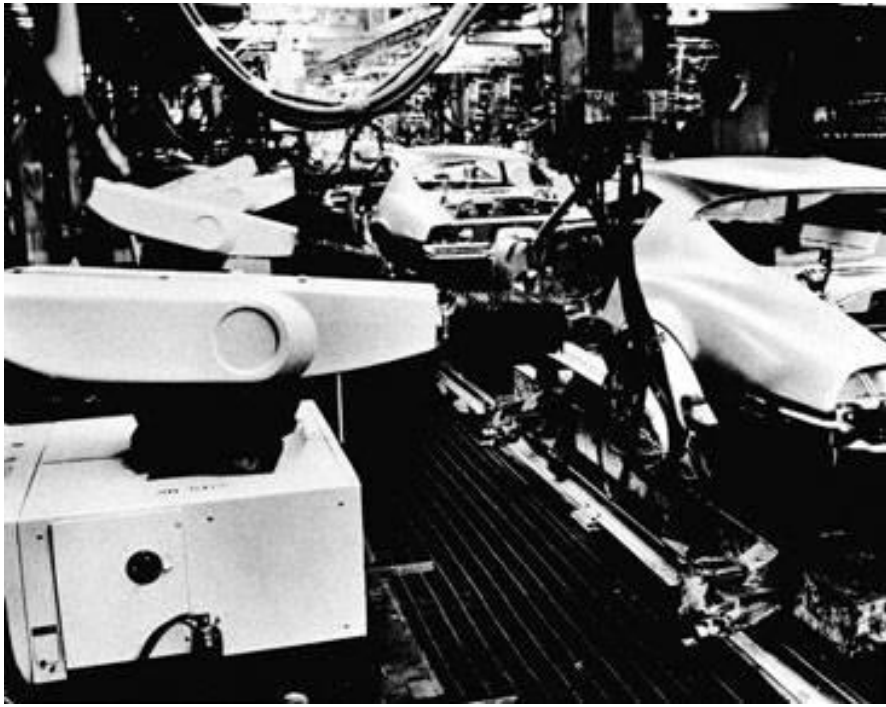
The origins of modern robotics



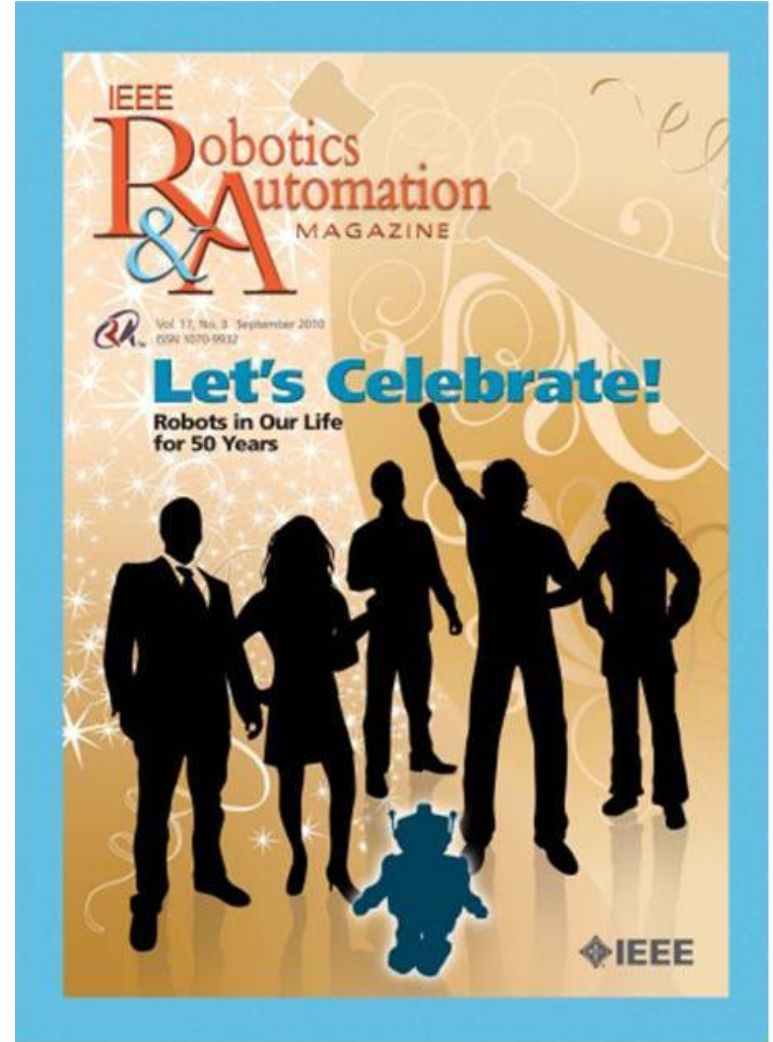
In 1960 the first industrial robot was installed in a manufacturing plant of General Motors in New Jersey (USA)



The origins of modern robotics



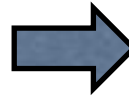
Robotics has grown exponentially in the last 65 years and the theories and techniques for robot control, fabrication and sensing represent an incredible wealth of knowledge. Robotics technologies are today very solid and robust, in the **accurate, fast, and reliable control of robot motion**



The origins of modern robotics

Industrial robotics:

birth and growth of theories and techniques for robot control



Service robotics:

birth and growth of theories and techniques for robot perception & action control

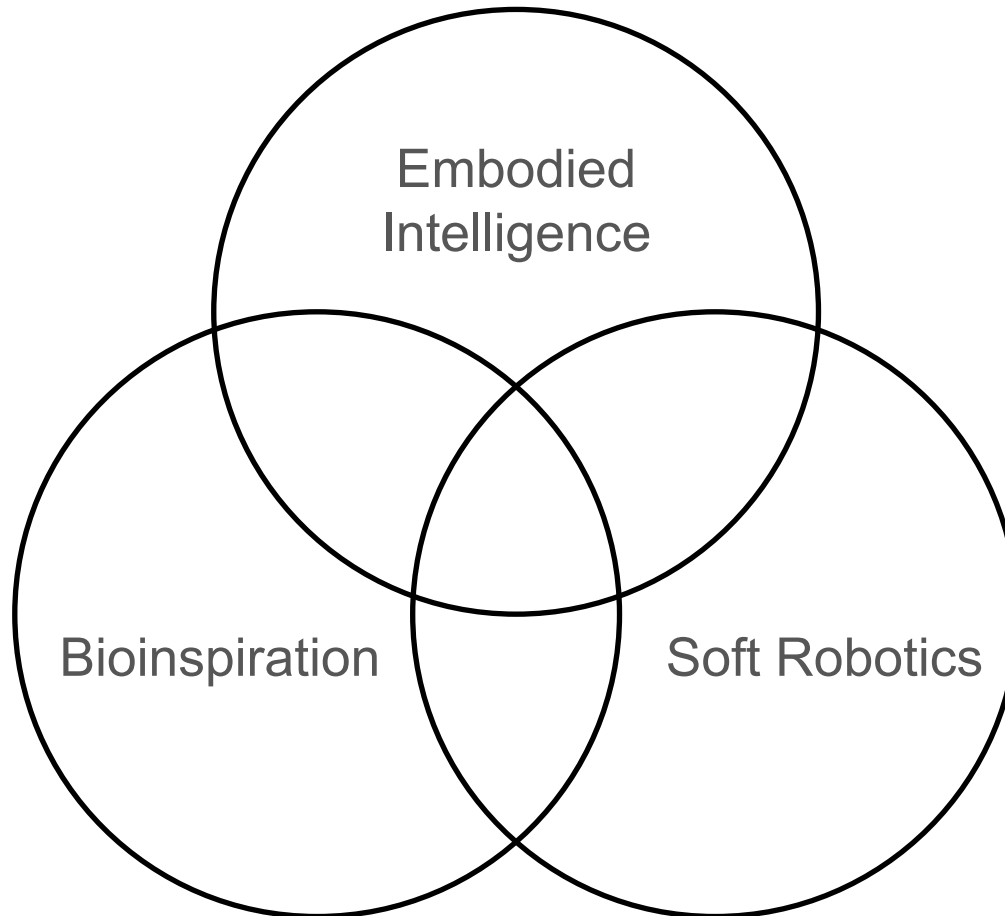
Structured environment



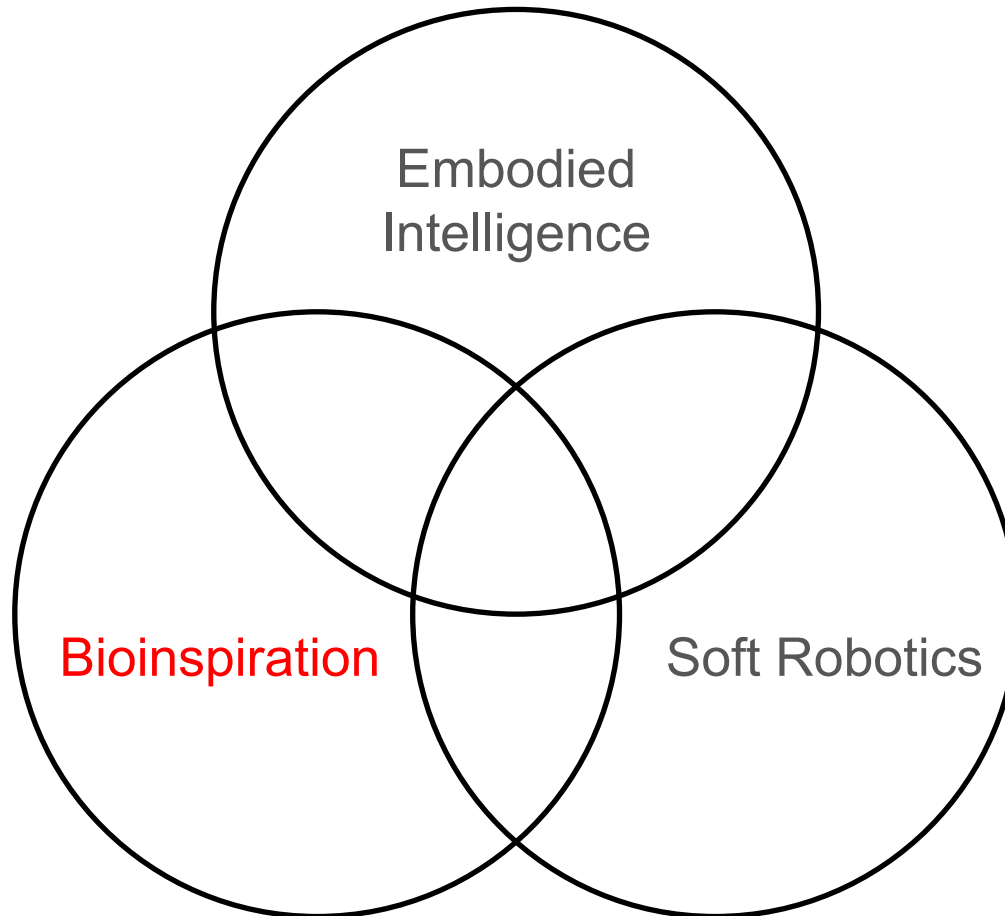
Unstructured environment



How to leave the cage?



How to leave the cage?



Abilities not yet reached by robots



Lessons from Nature: bioinspiration and biomimetics



Bioinspiration and biomimetics

Lessons from Nature: simplifying principles



Too simple?
Rather too complex?

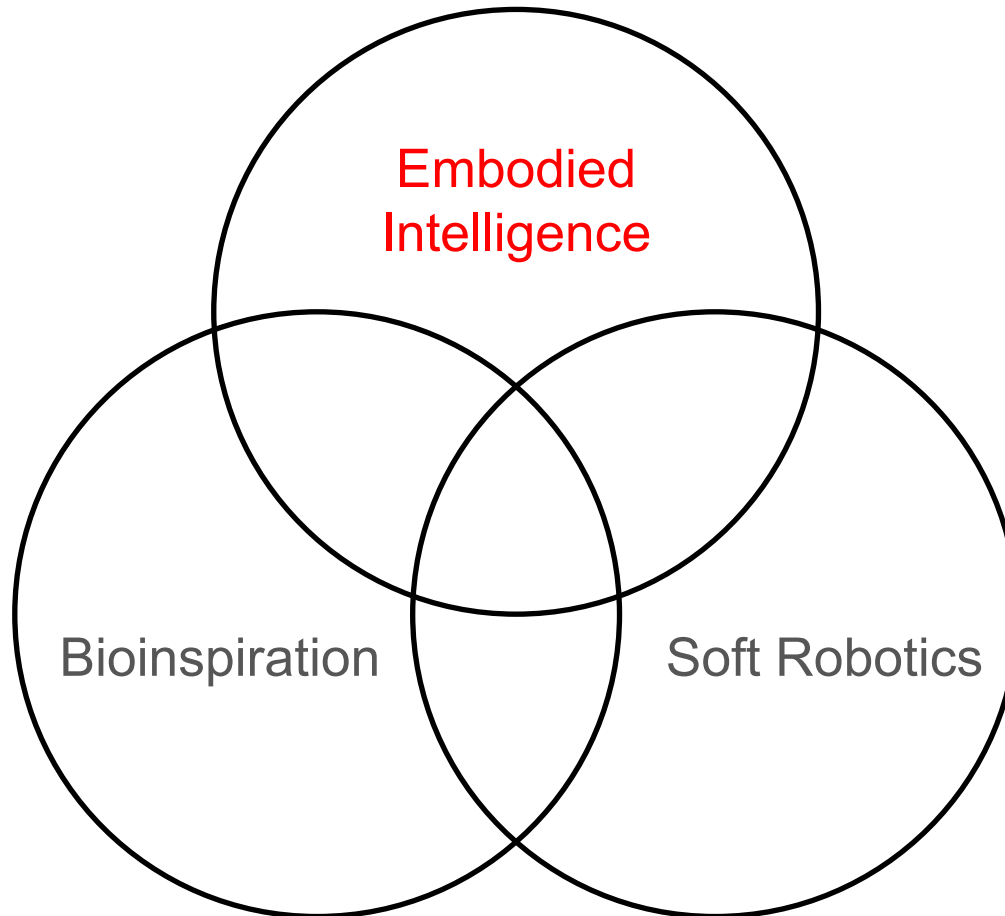


Studying living organisms and
understanding what makes their
behavior so smart and efficient

In robotics, we need **simplifying principles** for control and behavior



How to leave the cage?



Embodied Intelligence: the modern view of Artificial Intelligence

Classical approach

The focus is on the brain and central processing



Modern approach

The focus is on interaction with the environment. Cognition is emergent from system-environment interaction



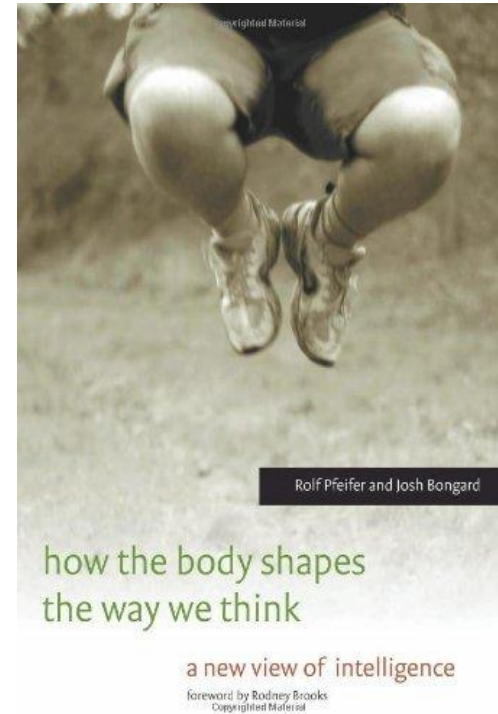
Rolf Pfeifer and Josh C. Bongard, How the body shapes the way we think: a new view of intelligence, The MIT Press, Cambridge, MA, 2007



Embodied Intelligence: the modern view of Artificial Intelligence

Adaptive behaviour is not just control and computation, but it emerges from the complex and dynamic interaction between the **morphology** of the body, **sensory-motor control**, and **environment**.

Many tasks become much easier if embodiment is taken into account.



The nature of the human mind is largely determined by the form of the human body

Rolf Pfeifer and Josh C. Bongard, *How the body shapes the way we think: a new view of intelligence*, The MIT Press, Cambridge, MA, 2007



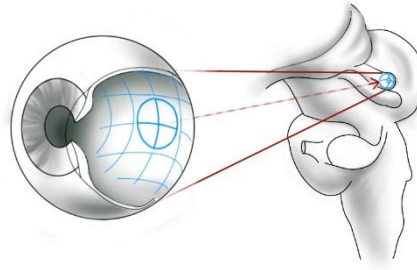
Morphological computation

As any transformation of information can be named as *computing*, *Morphological Computation* endows all those behaviours where computing is mediated by the properties of the physical body



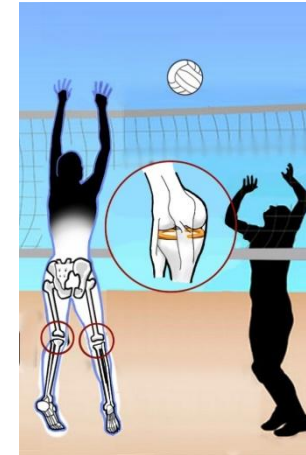
Shape

as body structure,
specifies the behavioral
response of the agent



Arrangement

of the motor,
perceptive and
processing units



Mechanical properties

allow emergent behaviors and
adaptive interaction with the
environment

Zambrano D, Cianchetti M, Laschi C (2014) "The Morphological Computation Principles as a New Paradigm for Robotic Design" in *Opinions and Outlooks on Morphological Computation*, H. Hauser, R. M. Fuchslin, R. Pfeifer (Ed.s), pp. 214-225.



Morphological computation

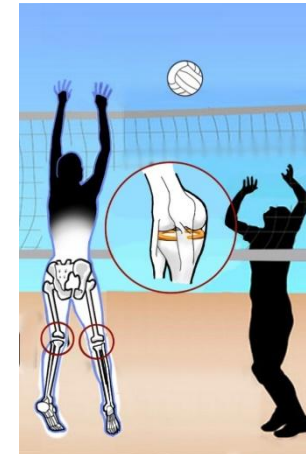
As any transformation of information can be named as *computing*, *Morphological Computation* endows all those behaviours where computing is mediated by the properties of the physical body

In order to grasp the simplifying principles of morphological computation, we need to

- exploit the mechanical feedback from the interaction with the environment
- use body compliance (instead of fighting it)

New bodyware

Soft bodyware?



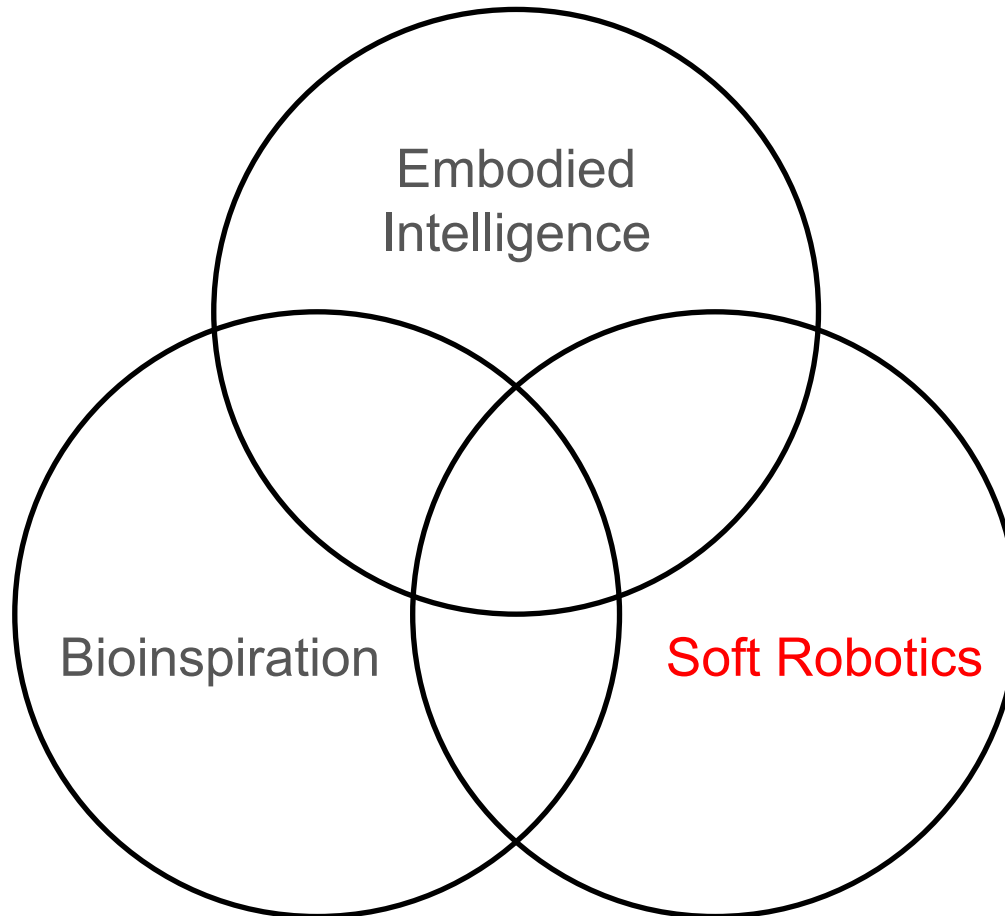
Mechanical properties

allow emergent behaviors and adaptive interaction with the environment

Zambrano D, Cianchetti M, Laschi C (2014) "The Morphological Computation Principles as a New Paradigm for Robotic Design" in *Opinions and Outlooks on Morphological Computation*, H. Hauser, R. M. Fuchslin, R. Pfeifer (Ed.s), pp. 214-225.



How to leave the cage?



Why use soft materials

- Richer behaviour
- Safer
- Mostly programmable



Why use soft materials not to

- Richer behaviour
- Safer
- Mostly programmable
- Hard to manage/design
- Variable stiffness needed
- Missing references



Soft Robotics

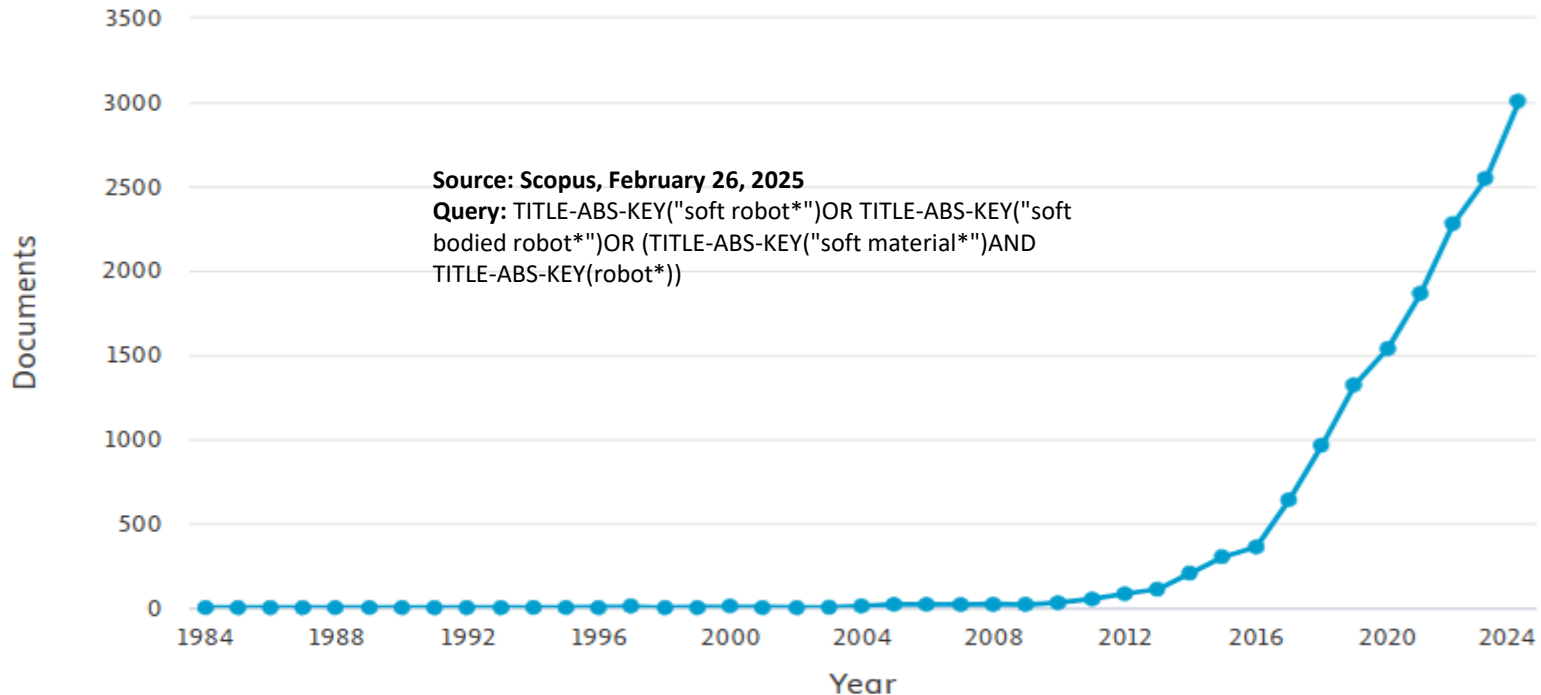
nature
International weekly journal of science

Science AAAS

Science Robotics AAAS



Documents by year



IEEE PULSE
MAY 2016
A MAGAZINE OF THE IEEE ENGINEERING IN MEDICINE AND BIOLOGY SOCIETY

frontiers
in Robotics
and AI
Soft Robotics

**BIOINSPIRATION
& BIOMIMETICS**
Learning from nature

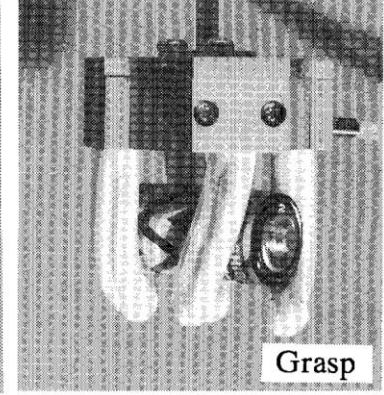
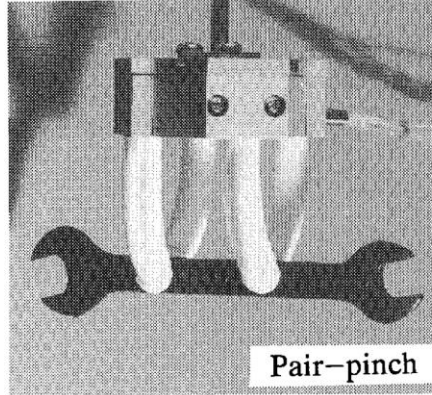
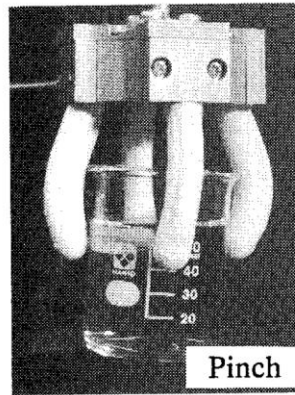
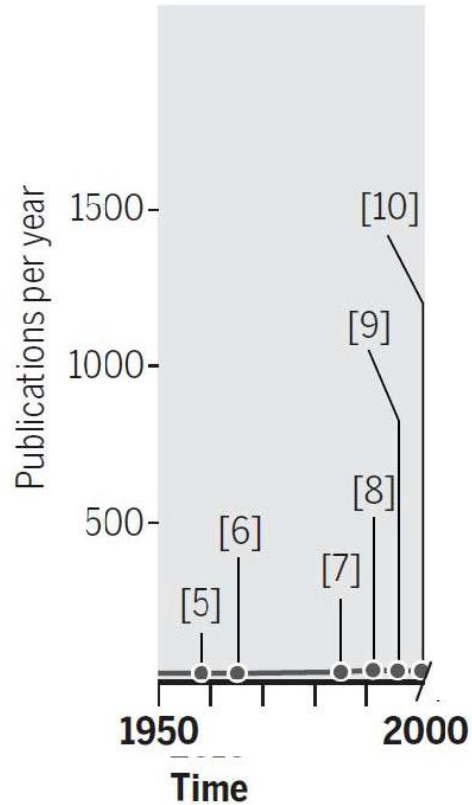
Soft Robotics
SoRo

IEEE SPECTRUM
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Soft Robotics: early discoveries

Early discovery



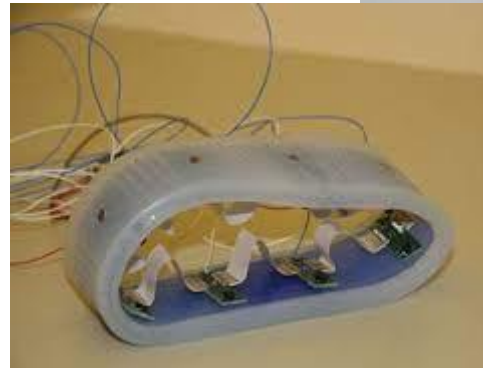
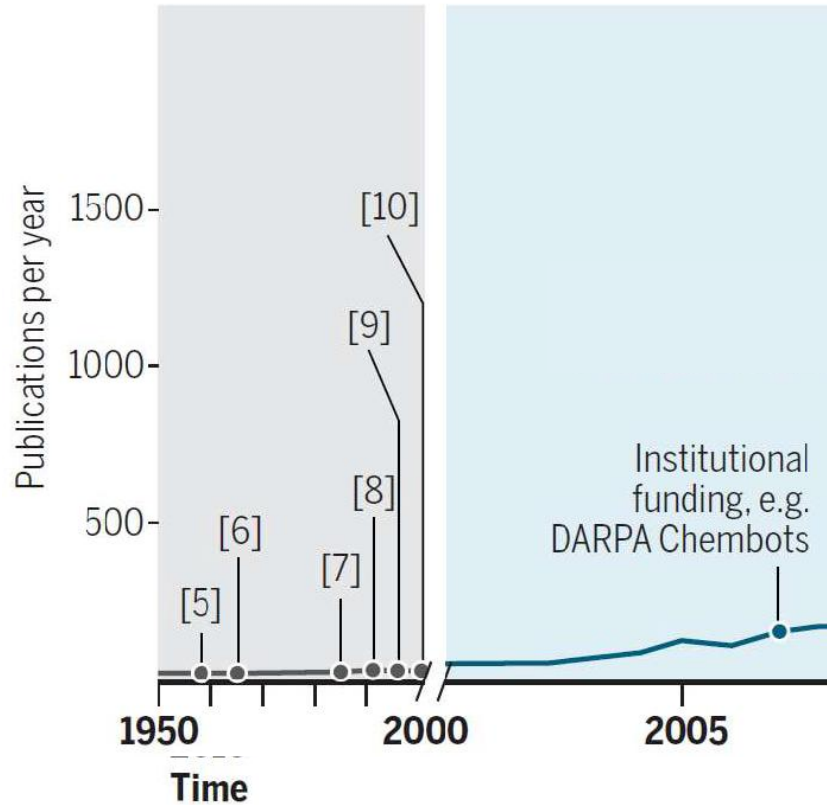
K SUZUMORI, S IIKURA and H TANAKA, Development of Flexible Microactuator and Its Applications to Robotic Mechanisms, ICRA 1989, pp.1622-1627.

Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", Science Robotics, 6(53).



Soft Robotics: the origin

Early discovery

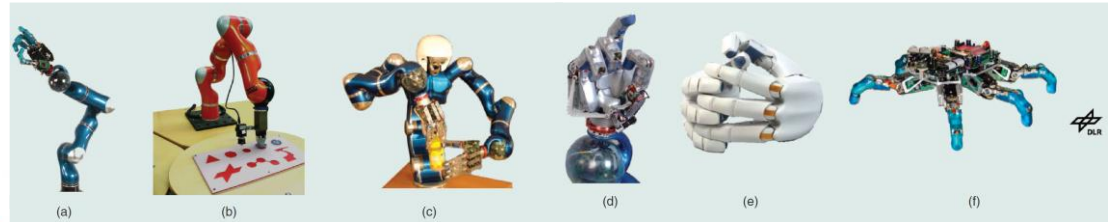


Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", *Science Robotics*, 6(53).



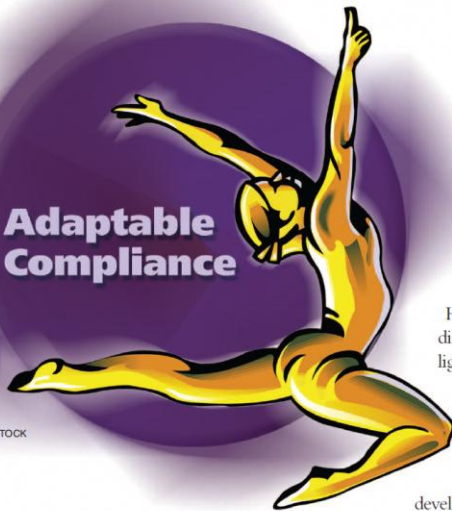
Soft Robotics: the origin

Albu-Schaffer A. et al.



Soft Robotics

Adaptable Compliance



©PUNCHSTOCK

After decades of intensive research, it seems that we are getting closer to the time when robots will finally leave the cages of industrial robotic workcells and start working in the vicinity of and together with humans. This opinion is not only shared by many robotics researchers but also by the leading automotive and IT companies and, of course, by some clear-sighted industrial robot manufacturers. Several technologies

required for a necessary level of performance in communication technology. However, it is clear that this is different than today's industrial robots, which are designed for different tasks, lightweight design, and so on. The expected performance of humans or in unknown environments present and compared to the mentioned soft-robot technology of torque

developed during the past decade at the German Aerospace Center (DLR) (arms, hands, a humanoid upper body and a crawler). Several products

From Torque Feedback-Controlled Lightweight Robots to Intrinsically Compliant Systems

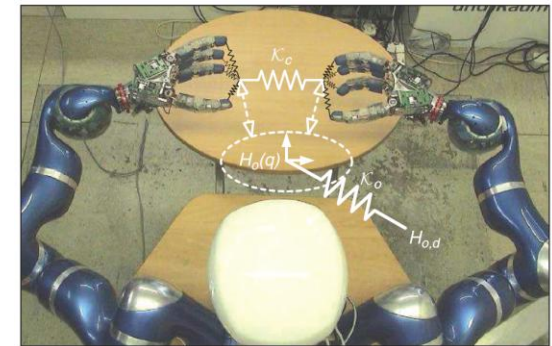


Figure 5. Two-hand impedance behavior by combining the object-level impedances of the hands and the arms.

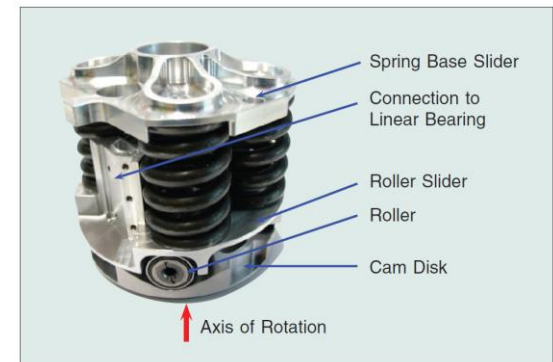


Figure 14. VS-joint mechanism. The link axis is in the vertical direction. The cam disk rotates on a compliant link deflection.

A. Albu-Schaffer et al., "Soft robotics," in IEEE Robotics & Automation Magazine, vol. 15, no. 3, pp. 20-30, September 2008.



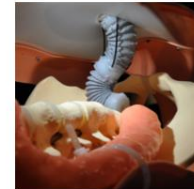
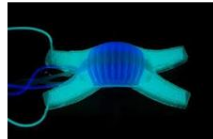
Soft Robotics: the origin

Soft Robotics: a working definition

Robot

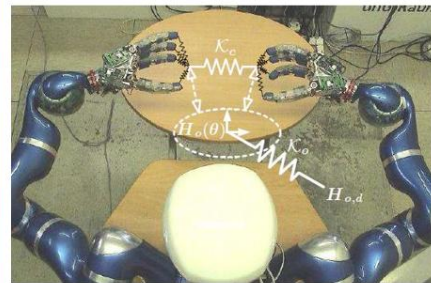
Use of soft materials in robotics

- * Robots made of soft materials that undergo large deformations during interaction
- * Soft actuators and soft components
- * Control partially embedded in the robot morphology and mechanical properties



Variable impedance actuators and stiffness control

- * Highly flexible (hyper-redundant or continuum) robots
- * High number of rigid links, multi-axis joints



IEEE Robotics and Automation Magazine, Special Issue on Soft Robotics, 2008

Presented at ICRA 2014



Soft Robotics: the origin

The definition "war"

Kim S., Laschi C., Trimmer B.



“Soft bodied robots”, in analogy with soft bodied animals

Kim S., Laschi C., and Trimmer B. (2013) Soft robotics: a bioinspired evolution in robotics, Trends in Biotechnology

Laschi C., and Cianchetti M.

“Robots built with soft materials”

Laschi C. and Cianchetti M. (2014) “Soft Robotics: new perspectives for robot bodyware and control” Frontiers in Bioengineering & Biotechnology , 2(3)

Wang L., Iida F.

“soft matter robotics”, using the concept of “soft matter” used for materials

L. Wang, F. Iida, Deformation in Soft Matter Robotics: A Categorization and Quantitative Characterization. IEEE Robotics & Automation Magazine 22(3), 125 139 (2015).



Soft Robotics: the origin

The definition "war"

Rus D., Tolley M.T.

“systems that are capable of autonomous behavior, and that are primarily composed of materials with moduli in the range of that of soft biological materials”

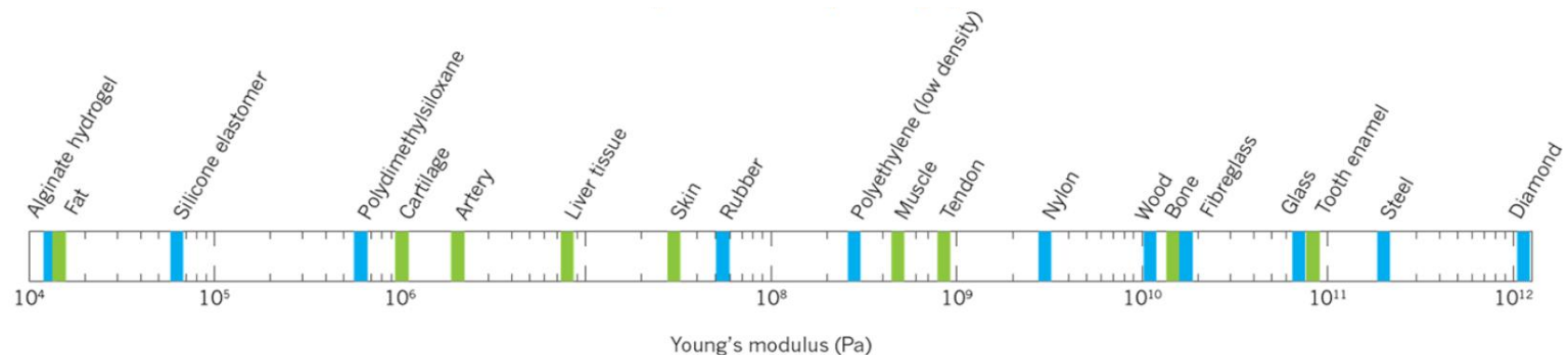


Figure 2 | Approximate tensile modulus (Young's modulus) of selected engineering and biological materials. Soft robots are composed primarily of materials with moduli comparable with those of soft biological materials (muscles, skin, cartilage, and so on), or of less than around 1 gigapascal. These materials exhibit considerable compliance under normal loading conditions.

D. Rus , M. T. Tolley, *Design, fabrication and control of soft robots. Nature* 521, 467 475 (2015)



Soft Robotics: the origin

The definition "war"

RoboSoft Community



First RoboSoft Working Paper – September 2014

On the basis of the above statements, the RoboSoft community proposed and agreed on the following definition of Soft Robotics:

“Soft robot/devices that can actively interact with the environment and can undergo ‘large’ deformations relying on inherent or structural compliance”

RoboSoft was a Coordination Action on Soft Robotics funded by the European Commission



Soft Robotics: the origin

The definition "war"



First RoboSoft Working Paper – September 2014

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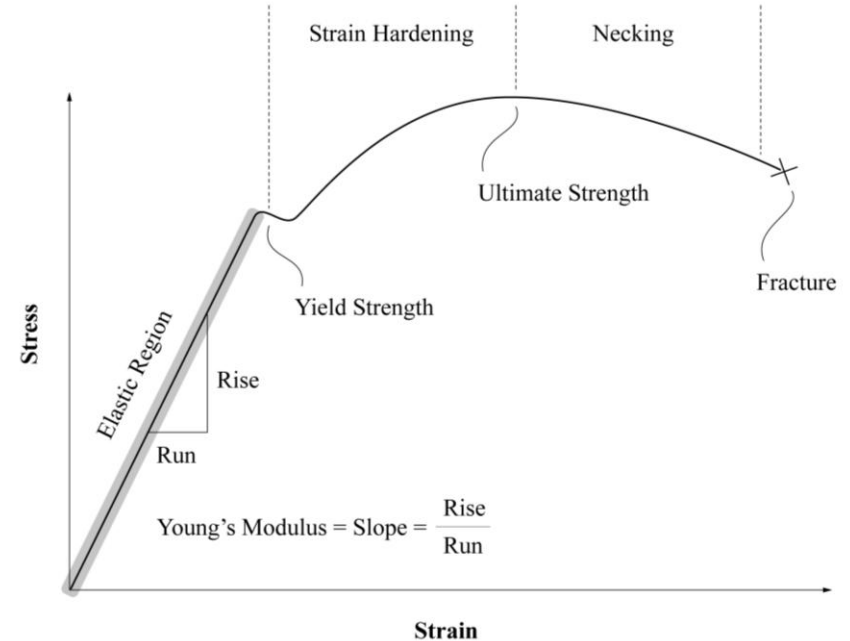
RoboSoft is a Coordination Action on Soft Robotics funded by the European Commission

$$k^{-1} \propto \frac{L}{EA}$$

LOW Elastic Modulus → Soft Robotics

HIGH Elastic Modulus → Hard Robotics

Geometry



$$E = \frac{\sigma}{\varepsilon}$$

expressed in Pascals (Pa)

E : Young's modulus

σ : stress

ε : strain



Soft Robotics: the origin

The definition "war"



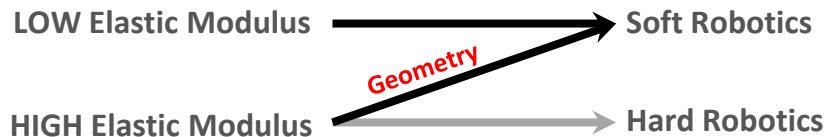
First RoboSoft Working Paper – September 2014

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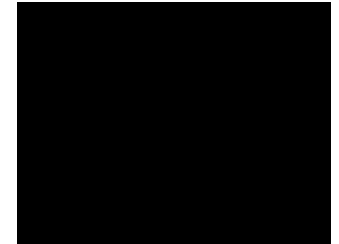
RoboSoft is a Coordination Action on Soft Robotics funded by the European Commission

$$k^{-1} \propto \frac{L}{EA}$$



Soft Robotics may use materials which present:

- INHERENT MATERIAL compliance: bulk material properties (elastomers, low elastic modulus polymers, gels...)



M. Wehner, R.L. Truby, D.J. Fitzgerald, B. Mosadegh, G.M. Whitesides, J.A. Lewis, R.J. Wood, An integrated design and fabrication strategy for entirely soft, autonomous robots, *Nature* 536, 451–455

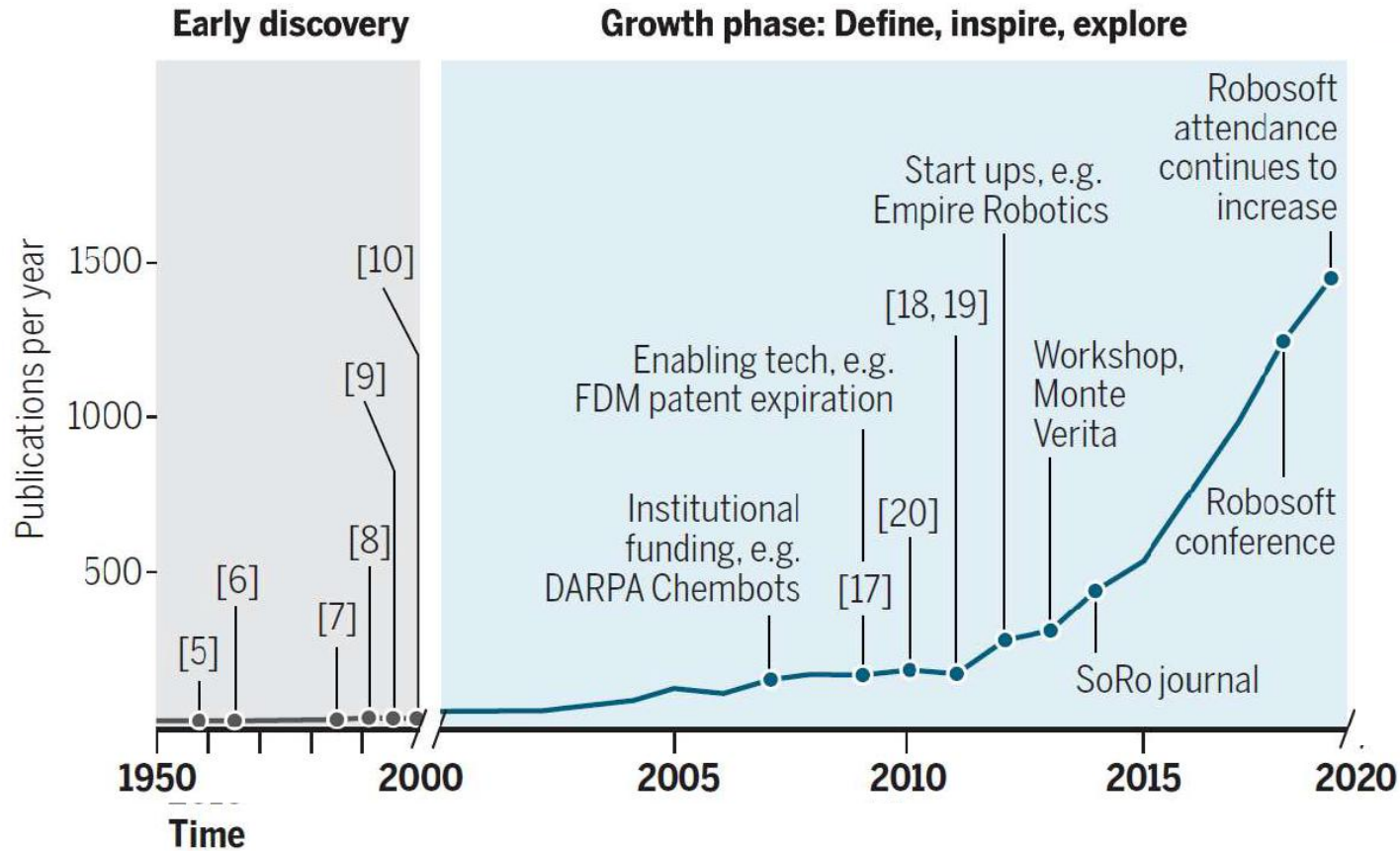
- STRUCTURAL compliance: geometric features or arrangement amplify strains with respect to local material deformation



Follador M., Cianchetti M., Arienti A., Laschi C., (2012) "A general method for the design and fabrication of shape memory alloy active spring actuators", *Smart Materials and Structures*



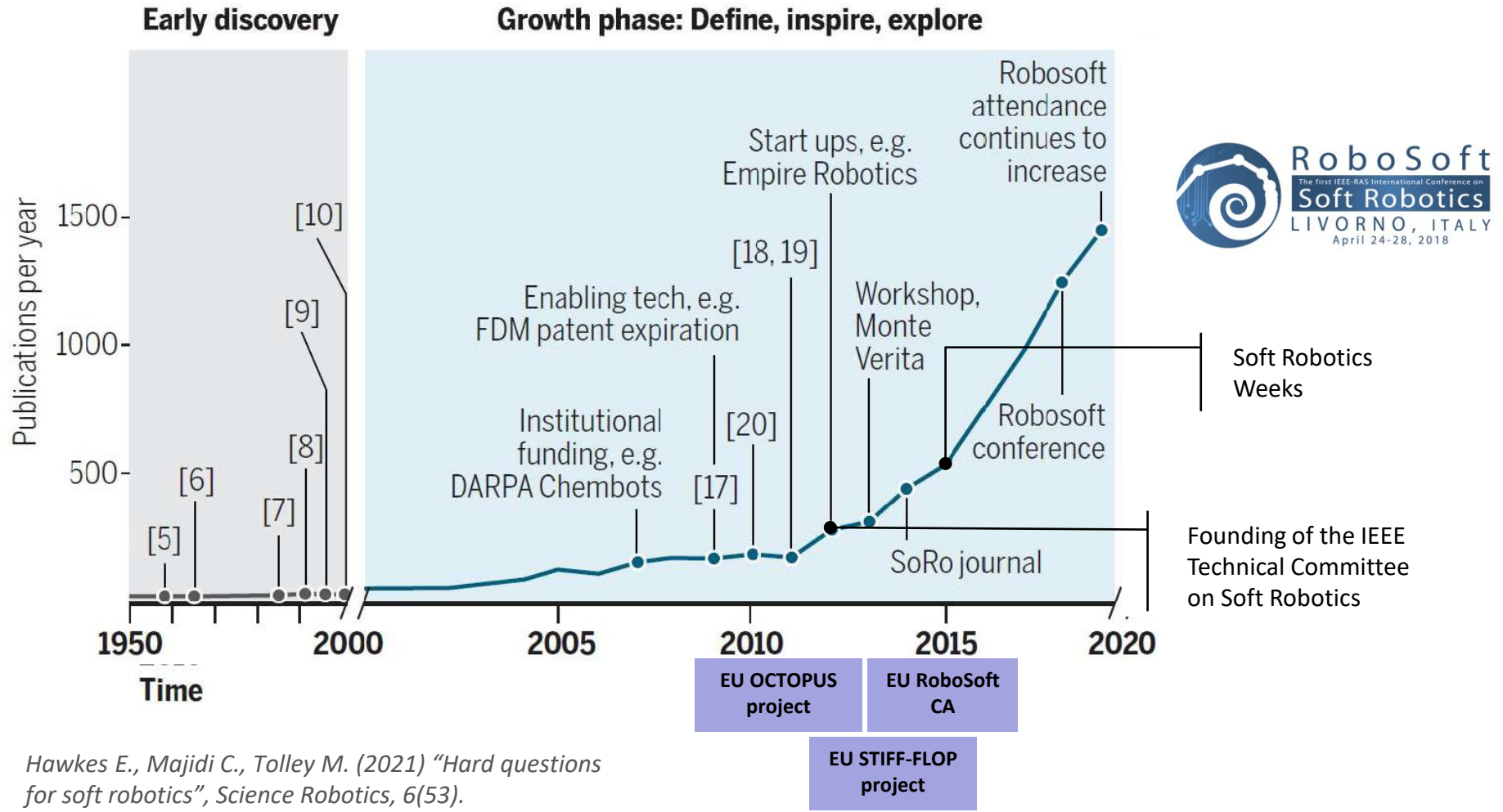
Soft Robotics: the growth



Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", *Science Robotics*, 6(53).



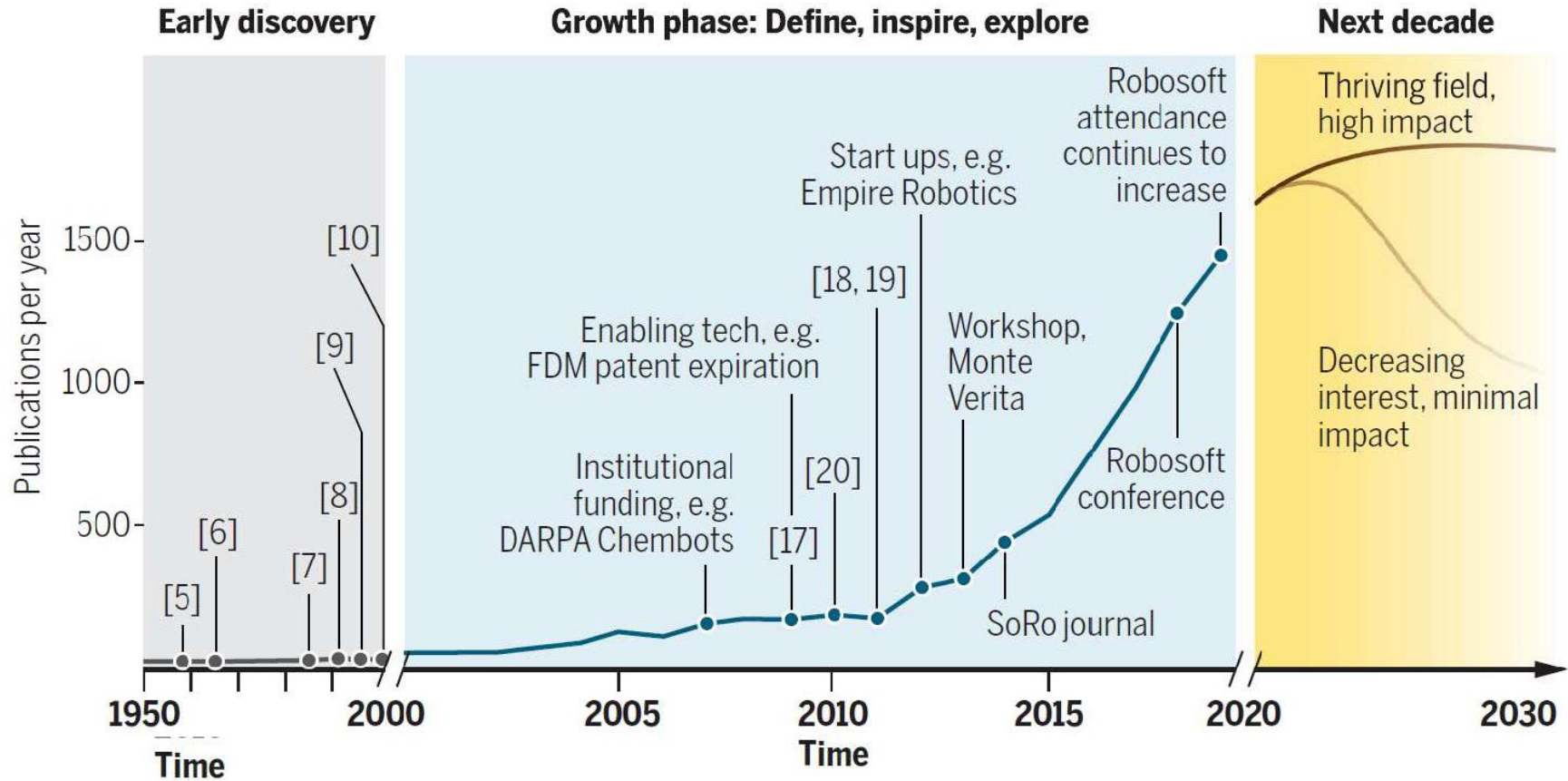
Soft Robotics: the growth



Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", *Science Robotics*, 6(53).



Soft Robotics: the crossroad



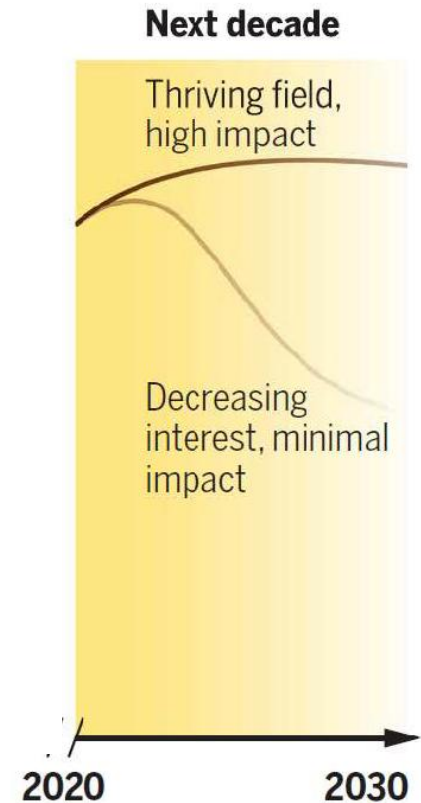
Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", *Science Robotics*, 6(53).



Soft Robotics: the crossroad

LEVELS OF CONTRIBUTION

- Level 0 (minimal contribution, limited advancement to the field in the second decade) New systems, similar performance that is achieved in a different manner.
- Level 1 (contribution within the field of soft robotics) Advances in soft robotics but limited impact on other research fields or technology domains.
- Level 2 (contribution to robotics and engineering more broadly) Not only moving forward soft robotics but also advancing other scientific fields and application domains more broadly



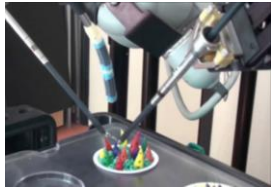
Hawkes E., Majidi C., Tolley M. (2021) "Hard questions for soft robotics", *Science Robotics*, 6(53).



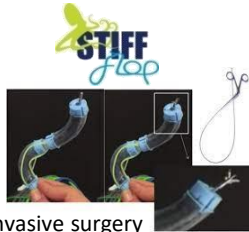
Soft Mechatronics for Biorobotics (SMB) Lab

From enabling technologies for soft actuators, smart compliant sensors and flexible mechanisms to full soft robotic systems based on soft/flexible materials

Surgical applications



Soft manipulators for minimally invasive surgery



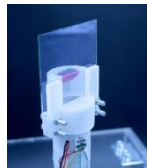
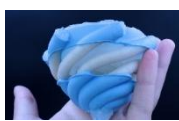
Assistive robotics

Manipulators and garments for personal assistance

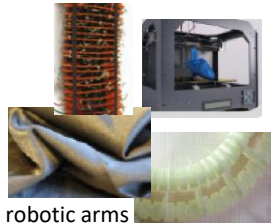


Simulators of body parts and artificial organs

Fully soft total artificial heart



Collaborative robotics



Soft robotic arms with bodily awareness



Soft robotic gripper for delicate manipulation and harvesting



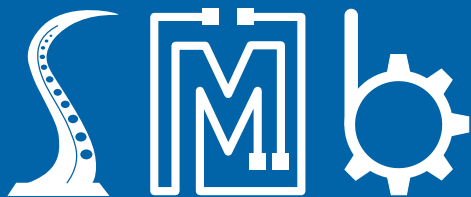
Food industry



Neurodevelopmental Engineering



Soft robotic devices for monitoring human motor capabilities



SOFT MECHATRONICS FOR BIOROBOTICS

Currently 21 members

